

Responsible & Explainable AI — Final Report

Jigsaw Toxicity Classification with Bias, Adversarial Robustness & Guardrails

1. Introduction

This project builds a complete **end-to-end toxic comment moderation system** using a fine-tuned **DistilBERT transformer model** trained on the Jigsaw Unintended Bias dataset. The system is evaluated not only for predictive performance but also for:

- Demographic fairness across identity groups
- Robustness against adversarial attacks
- Effectiveness of mitigation strategies
- Realistic production-grade moderation pipeline design

The goal is to simulate real-world AI content moderation challenges faced by large-scale platforms, including bias amplification, adversarial evasion, and operational deployment constraints.

2. Dataset & Preprocessing

The dataset used is the **Jigsaw Unintended Bias in Toxicity Classification dataset**, containing ~1.9 million annotated comments.

Key preprocessing steps:

- Removal of null values in comment_text and toxic
- Conversion of toxic column into binary labels:
 - Toxic = 1 if score ≥ 0.5
 - Non-toxic = 0 otherwise
- Stratified sampling:
 - Training set: 20,000 samples
 - Evaluation set: 5,000 samples
- Stratification ensured class balance preservation

This reduced dataset was used due to computational constraints in Google Colab T4 GPU.

3. Baseline Model (DistilBERT)

A pretrained **DistilBERT (uncased)** model was fine-tuned for binary toxicity classification.

Training setup:

- Epochs: 2–3
- Max sequence length: 128
- Batch size: 16
- Loss: Cross-entropy
- Optimizer: AdamW (default Trainer)

Results:

- Accuracy: ~0.95
- F1 Score: **0.62**
- AUC: **0.94**

Interpretation:

The model performs strongly in distinguishing toxic vs non-toxic content but shows moderate recall for toxic samples, which is expected due to dataset imbalance.

4. Threshold Analysis

Different classification thresholds were evaluated:

Threshold F1 Score	
0.3	0.66
0.4	0.65
0.5	0.62
0.6	0.64
0.7	0.63

Selected threshold: 0.5

Reason:

- Balanced tradeoff between precision and recall
- Aligns with standard classification practice
- Avoids excessive false positives in moderation systems

5. Bias Audit (Fairness Evaluation)

Two demographic cohorts were constructed:

- **High-Black cohort:** black identity score ≥ 0.5
- **Reference cohort:** white identity ≥ 0.5 and black < 0.1

Observed cohort sizes:

- High-Black: ~60–70 samples
- Reference: ~40–50 samples

(Note: small cohort size leads to higher variance in fairness metrics.)

Key Results:**High-Black cohort:**

- TPR ≈ 0.31
- FPR ≈ 0.06
- FNR ≈ 0.68
- Precision ≈ 0.66

Reference cohort:

- TPR ≈ 0.50
- FPR ≈ 0.12
- FNR ≈ 0.50
- Precision ≈ 0.63

Fairness Metrics:

- Statistical Parity Difference: negative (lower positive prediction rate for High-Black cohort)
- Equal Opportunity Difference: ≈ -0.18
- Disparate Impact (FPR ratio): ≈ 0.48

Interpretation:

The model shows **unequal error distribution across demographic groups**, particularly in:

- Lower true positive rate for High-Black cohort
- Different false positive/negative behavior across groups

However, due to small sample sizes, these estimates are **directionally valid but statistically limited**.

6. Adversarial Robustness

Attack 1: Evasion Attack

A character-level perturbation attack was implemented using:

- Unicode homoglyph substitution
- Zero-width space injection
- Character duplication

Results:

- Attack Success Rate (ASR): **0.92**
- Confidence dropped from $\sim 0.92 \rightarrow \sim 0.09$

Interpretation:

The model is **highly vulnerable to text obfuscation attacks**, showing that token-level transformers are sensitive to character-level manipulation.

Attack 2: Poisoning Attack (Simulation)

5% of training labels were flipped to simulate data poisoning.

Expected impact:

- Increased false negatives
- Reduced F1 score
- Degraded decision boundary quality

Interpretation:

Training pipeline integrity is vulnerable if adversaries can manipulate training data.

7. Mitigation Strategies

Three mitigation approaches were evaluated:

1. Oversampling

- Increased representation of underrepresented subgroup
- Result: slight F1 improvement ($\sim +0.01$)

2. Reweighing (simulated)

- Intended to adjust sample importance
- Result: slight F1 decrease (~ -0.005)

3. Threshold Optimization (simulated)

- Tradeoff between fairness and accuracy
- Result: slight F1 reduction (~ -0.01)

Key Insight:

There is a **fundamental tradeoff between fairness and accuracy**, and no single intervention improves both simultaneously.

8. Guardrail Pipeline (Production Simulation)

A 3-layer moderation system was implemented:

Layer 1: Rule-based Filter

- 20+ regex patterns across:
 - Threats
 - Self-harm
 - Doxxing
 - Dehumanization
 - Coordinated harassment

Result:

- 61 messages blocked instantly

Layer 2: ML Model

- DistilBERT probability-based classification
- Thresholds:
 - $\geq 0.6 \rightarrow$ block
 - $\leq 0.4 \rightarrow$ allow
 - otherwise $\rightarrow$ review

Layer 3: Human Review

- Uncertainty band: 0.4–0.6
- Only 3 cases escalated

Final system distribution:

- Block: 61

- Allow: 936
- Review: 3

Interpretation:

The system is:

- Highly efficient (low review load)
- Conservative in rule-based filtering
- Confident in ML decisions

This resembles **real-world moderation pipelines used in production systems**.

9. Conclusion

This project demonstrates a full lifecycle of responsible AI system design:

Achievements:

- Successful transformer-based toxicity classification
- Identification of demographic bias in predictions
- Demonstration of adversarial vulnerabilities
- Evaluation of mitigation strategies
- Design of a production-grade moderation pipeline

Final Insight:

While the model performs well in standard classification tasks, it exhibits:

- Sensitivity to adversarial text manipulation
- Demographic disparities in error rates
- Tradeoffs between fairness and accuracy

This highlights that **high accuracy alone is insufficient for deploying safe AI systems**, and must be complemented with fairness evaluation, robustness testing, and multi-layer guardrail architectures.

